ARMORER'S BENCH TECHNICAL MANUAL

Ruger 10/22 Carbine & Takedown

Ruger • The World's Most Popular Semi-Automatic Rimfire Rifle (.22 LR)

Manufacturer:	Ruger	**Category:**	Rifles & Carbines
Action Type:	Blowback Semi-Automatic Rimfire	**Era / Service:**	American Rimfire Legend (1964–Present)
Calibers:	.22 LR (Long Rifle)	**Covered Models:**	10/22 Carbine, 10/22 Takedown, 10/22 Target, 10/22 Sporter, Charger Pistol

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: V-Block Clamp Torque (25–30 in-lbs) • Rotary Magazine Pre-Tensioning (1-1/4 turns) • Action Screw Torque (15–20 in-lbs) • Auto-Bolt Release Modification • Rim Recess Headspace Stoning (0.0425")

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
V BLOCK SCREWS	25 to 30 in-lbs tightened in even alternating increments; overtightening causes barrel droop and accuracy loss
ACTION SCREW TORQUE	15 to 20 in-lbs (wood/synthetic stock) / 25 in-lbs (aluminum pillar bedding)
BOLT RIM HEADSPACE	0.0425" to 0.0435" true headspace cut into bolt face recess for match accuracy
MAGAZINE TENSION	Rotor clock spring pre-loaded exactly 1-1/4 full turns (450°) counter-clockwise before closing nut
TRIGGER PULL WEIGHT	6.0 to 7.0 lbs factory standard; drops to 2.25–2.75 lbs with Volquartsen or BX-Trigger

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Stovepipe / Failure to Eject on Last Rounds of Magazine **Spent rimfire case caught sideways in ejection port; occurs frequently as magazine empties**	• Factory stamped sheet metal extractor has rounded blunt claw hook. • Weak rotary magazine rotor spring tension failing to present final cartridges firmly. • Wax and carbon buildup inside bolt extractor pocket.	**Gunsmith Action: Install Volquartsen Exact Edge tool-steel wire-EDM sharp extractor and heavy spring. Clean bolt face extractor pocket. Disassemble 10-round rotary magazine and retension rotor spring to 1-1/4 turns.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Ruger 10/22 Complete Disassembly, Cleaning & Upgrades	Brownells Gunsmithing Bench	Teardown of the Ruger 10/22 action: V-block removal, bolt disassembly, auto-bolt release modification, and installing the BX trigger.